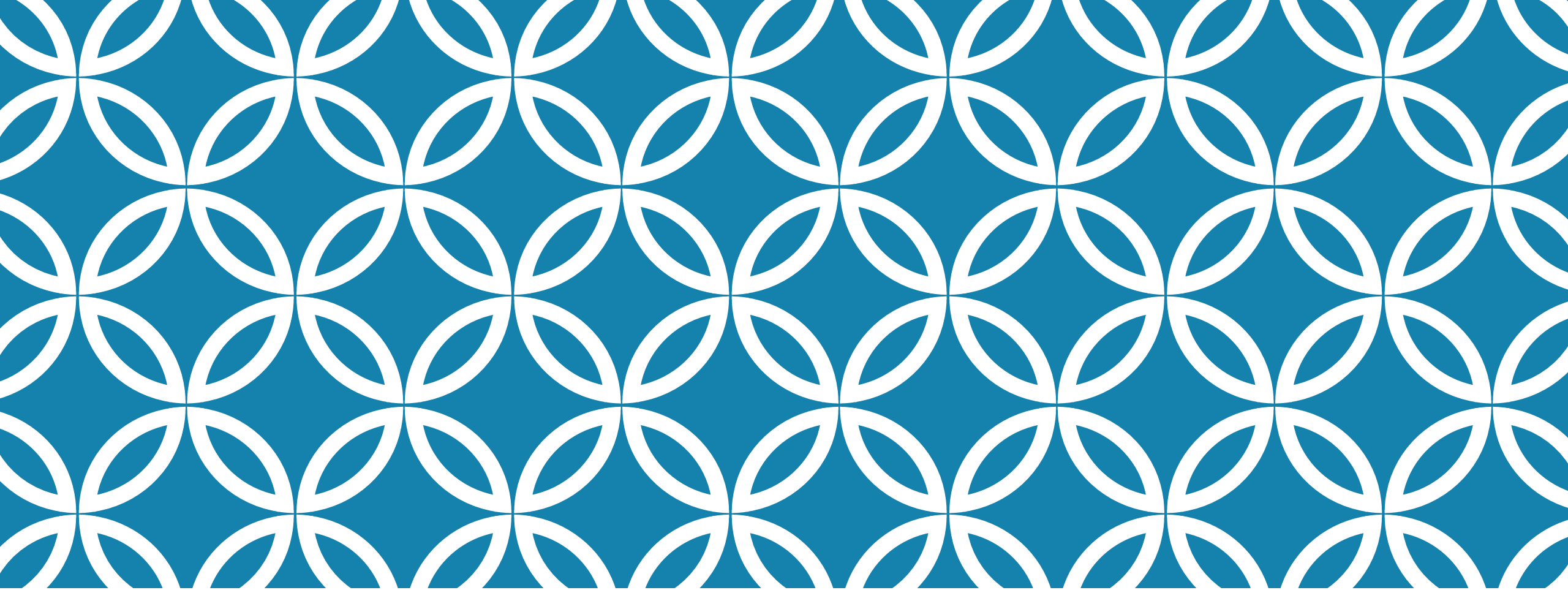


CLR PARSER

CLR(1) PARSER

CLR stands **for Canonical LR parser**

It uses **LR(1)** canonical items

Size of the **CLR parsing table is large** as compared to any other parsers

The 1 in CLR(1) is a lookahead item or symbol

LR(1) ➔ LR(0) + lookahead

EXAMPLE – 1

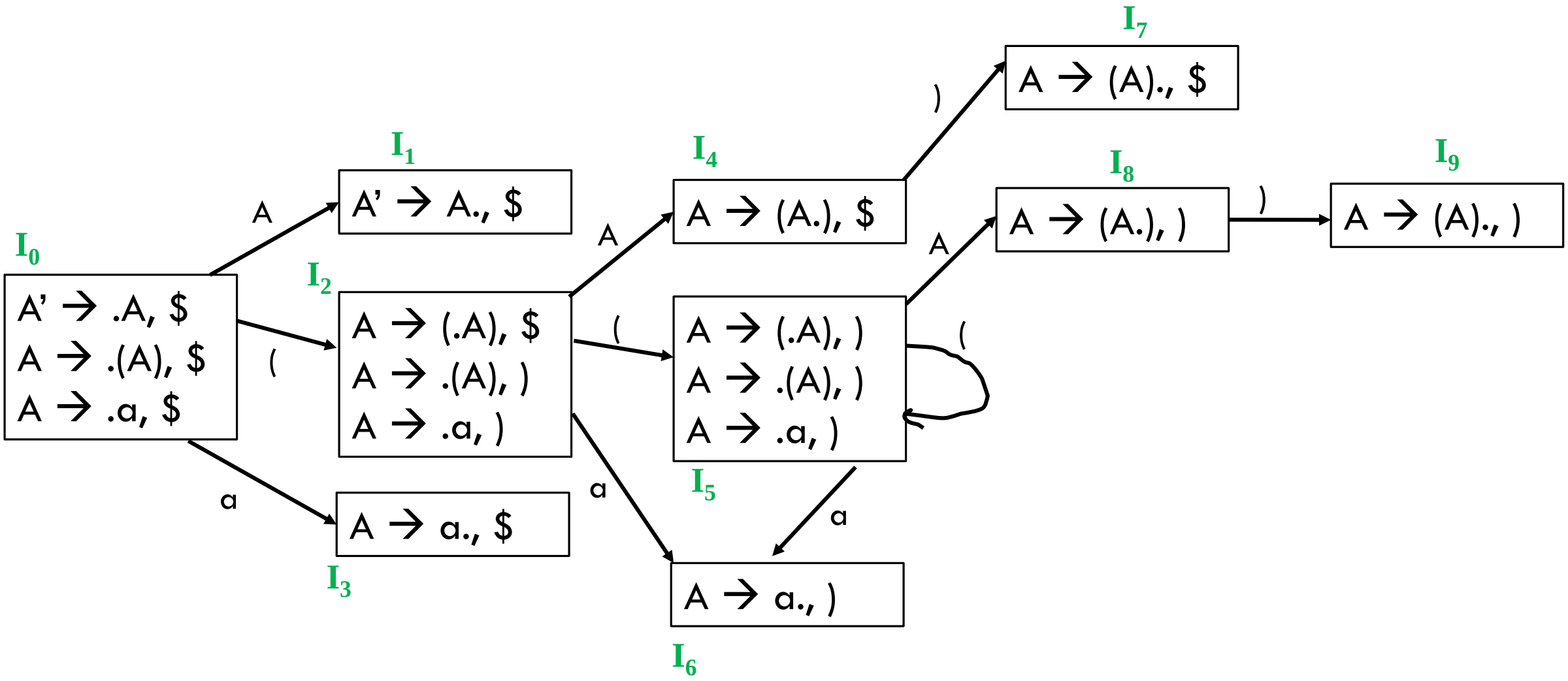
A → (A) | a

Let us find out whether this grammar can be parsed using CLR parser

0: A' → .A

1: A → .(A)

2: A → .a


I0
A' → .A, $
A → .(A), $
A → .a, $
A
(
a
I1
A' → A., $
I2
A → (.A), $
A → .(A),)
A → .a,)
A
(
a
I3
A → a., $
I4
A → (A.), $
I5
A → (.A),)
A → .(A),)
A → .a,)
A
(
a
I6
A → a.,)
)
I7
A → (A)., $
I8
A → (A.),)
)
I9
A → (A).,)

States	Action				Goto
	(	)	a	$	A
I0	S2		S3		1
I1				Accept	
I2	S5		S6		*4*
I3				R2	
I4		S7			
I5	S5		S6		8
I6		R2			
I7				R1	
I8		S9			
I9		R1			

Since there is no shift-reduce or reduce-reduce conflict, hence the grammar can be parsed using CLR parser

EXAMPLE – 2

S → AaAb | BbBa

A → ε

B → ε

Let us find out whether this grammar can be parsed using CLR parser

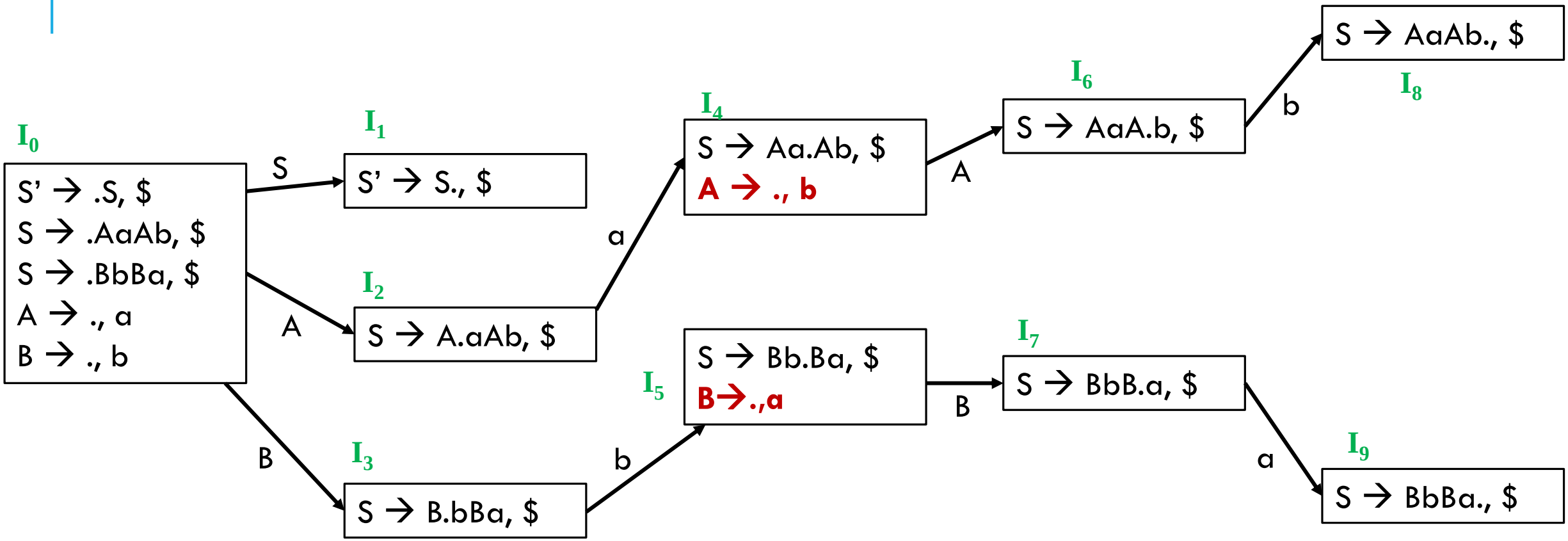
0: S' → .S

1: S → .AaAb

2: S → .BbBa

3: A → .

4: B → .


I0
S' → .S, $
S → .AaAb, $
S → .BbBa, $
A → ., a
B → ., b
S
A
B
I1
S' → S., $
I2
S → A.aAb, $
I3
S → B.bBa, $
a
b
I4
S → Aa.Ab, $
A → ., b
I5
S → Bb.Ba, $
B→.,a
A
B
I6
S → AaA.b, $
I7
S → BbB.a, $
b
a
I8
S → AaAb., $
I9
S → BbBa., $

States	Action			Goto		
	a	b	$	S	A	B
I0				1	2	3
I1			Accept			
I2	S4					
I3		S5				
I4		R3			6	
I5	R4					*7*
I6		S8				
I7	S9					
I8			R1			
I9			R2			

Since there is no shift-reduce or reduce-reduce conflict, hence the grammar can be parsed using CLR parser

EXERCISE

S → Aa | bAc | Bc | bBa

A → d

B → d

Let us find out whether this grammar can be parsed using CLR parser

Also check out exercises from the previous slides (LR(0) and SLR parser)